M5 Pulse

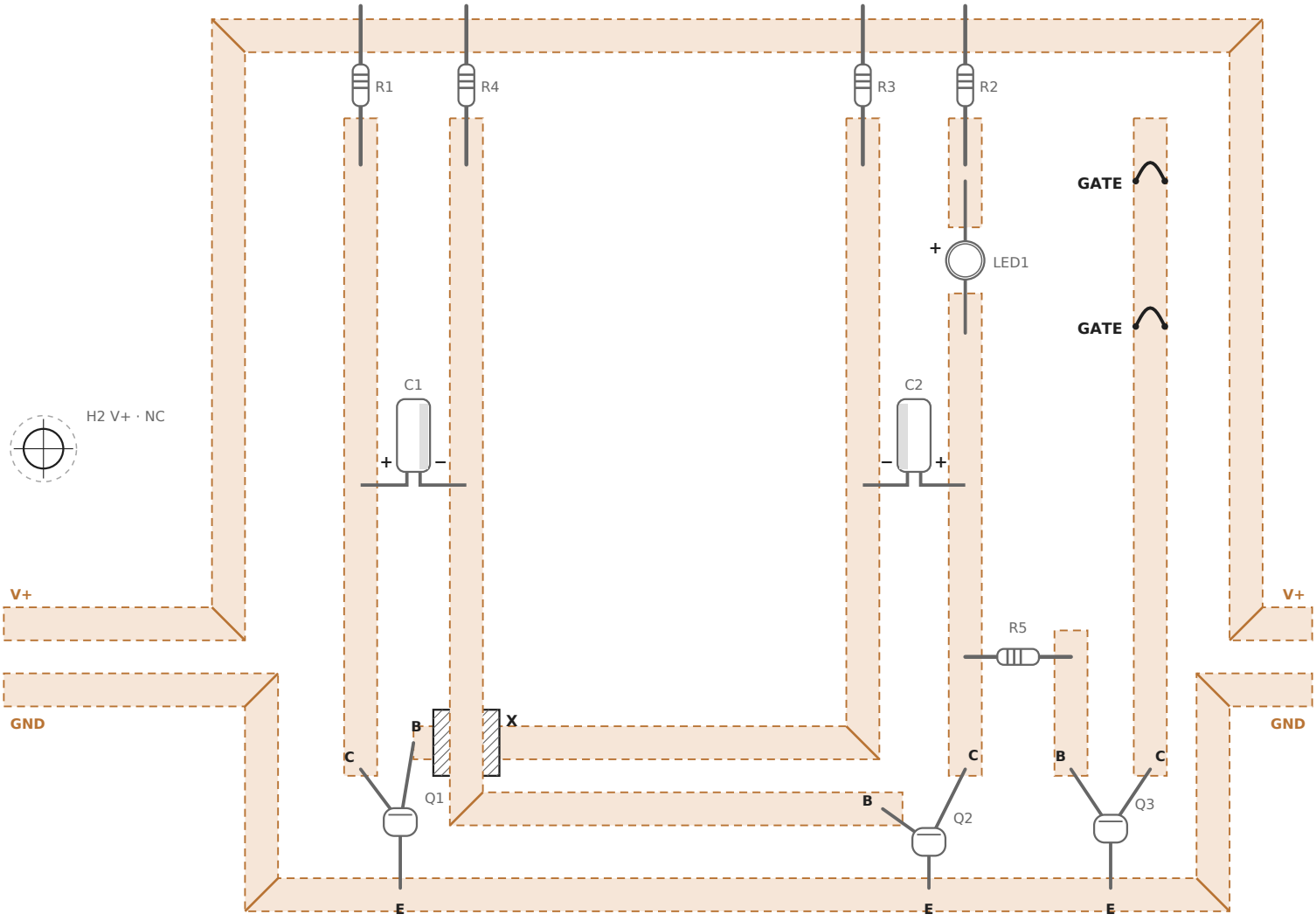
A slow oscillator blinks an LED and chops the note into a beat.

Draft: tagged steps may change

H1 BUS A · NC

You need

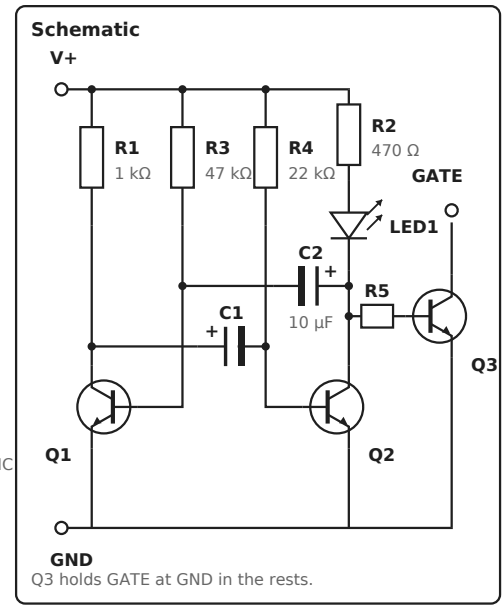
- Q1, Q2, Q3: 2N3904 transistors
- C1, C2: 10 µF electrolytic, 16 V or more
- R1 1 kΩ · R2 470 Ω · R3 47 kΩ
- R4 22 kΩ · R5 100 kΩ · LED1: red, 5 mm
- Bare wire for 2 patch loops
- Copper tape, solder, 2-core cable



H2 V+ · NC

H3 GND · NC

Patch one GATE to M1's PITCH and the controller's lead to the other. LED1 lights for each note.



H4 BUS B · NC

Build it

1. Lay the power ring, then the seven lanes inside it, one piece of tape each. Fold at the creases.
2. Where two lanes cross, lay the lower one, stick the insulating patch over it, then lay the upper one.
3. Place R1 to R5 either way round, and LED1 long leg on the +. C1 and C2 go + where printed: backwards, they can heat up and fail.
4. Lay Q1 to Q3 flat face up and bend each leg to its letter: E, B and C.
5. Solder every lead to its lane, or hold it flat with a small patch of copper tape. **[T0]**
6. Solder a patch loop upright on each GATE loop mark. **[T0]**
7. Join the edge connectors to the pages either side with 2-core cable, or with red and black clip leads.

Play it

- Power up: LED1 should blink about twice a second before anything is patched.
- Patch one GATE loop to M1's PITCH, and clip the controller's lead (M2's stylus, M3 or M4) to the other.
- For a slower beat, use 22 µF for C1 and C2. Swap R3 and R4 for short notes and long rests.

[T0] and **[P]** mark choices that may change: **[T0]** after the contact tests, **[P]** once the power supply is settled.

Print at 100 %: this bar should measure 100 mm.